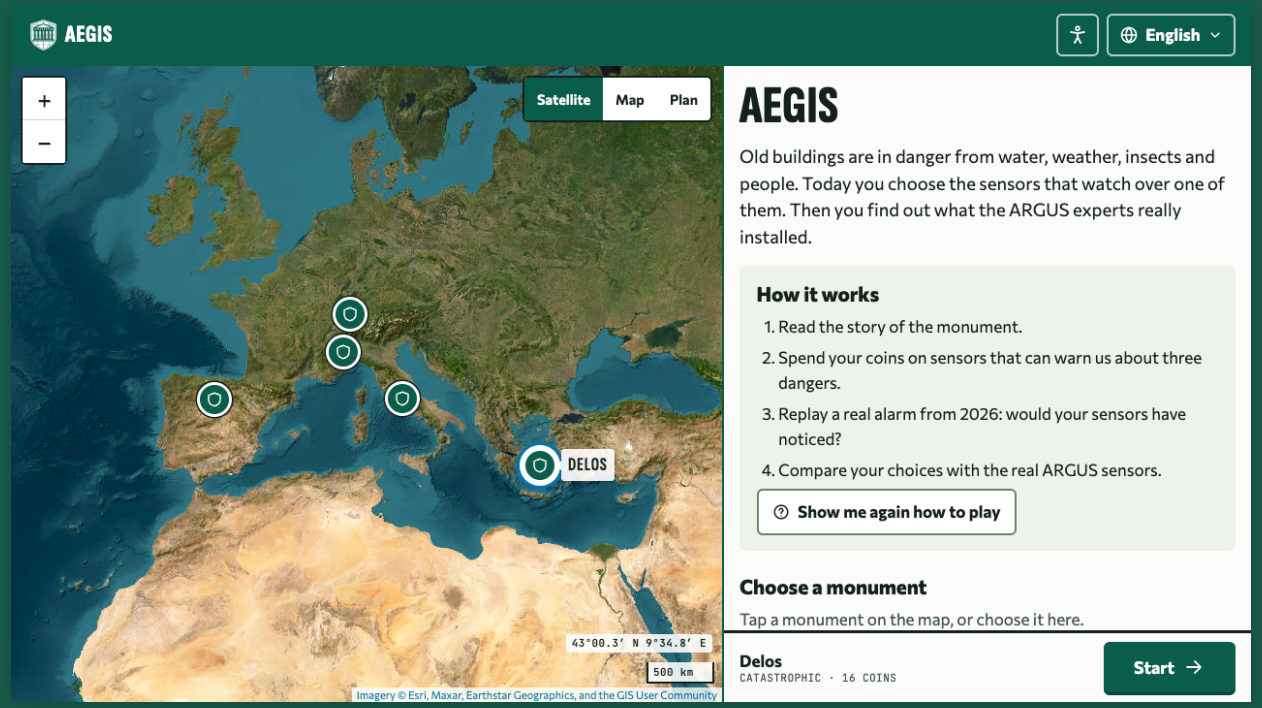
AEGIS

The visitor becomes a heritage guardian

ARGUS Open Innovation Challenge 2026
Track A: Heritage and Citizen Science



Play on your computer
or mobile:
`https://aloualou56.github.io/argus_trackA/`

The problem: making ARGUS data understandable

ARGUS monitors five European heritage sites, assesses their risks each day, and raises alarms when readings are unusual. When we examined its data, we found a gap between the technical records and what a visitor can understand.

An alarm as recorded by ARGUS


```
{"type": "DL",
 "score": 0.026571929454803467,
 "threshold": 0.020811407233135213,
 "is_alarm": true,
 "confidence_score": 0.5004581977196843,
 "top_feature_contributions": {
  "149368_h_masks_in_room215_2_soil_moisture": 0.4397,
  "151836_h_masks_in_room214_hum": 0.2579,
  "168439_h_masks_in_room215_lux": 0.0625, ...},
 "site": "delos",
 "timestamp": "2026-08-31T20:30:00"}
```


The same alarm in AEGIS

“On 31 August 2026, the ARGUS AI system noticed unusual soil moisture beneath a mosaic.”

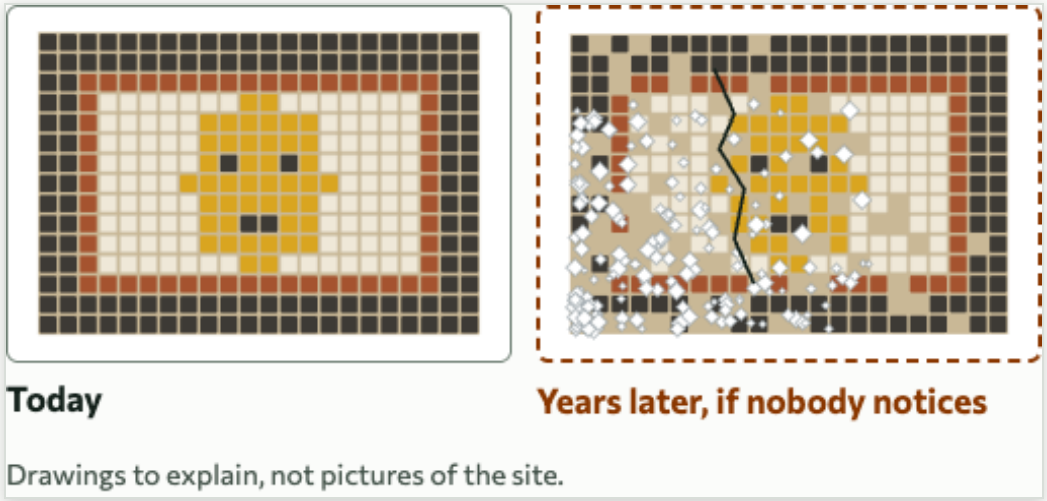
“This sensor accounts for 44% of the alarm, more than any other signal.”

Our approach uses experiential learning through a serious role-playing game to explain the work of ARGUS. The game uses authentic monitoring data, follows Kolb’s learning cycle, and extends into citizen science.

Left: ARGUS event record 123, with selected fields. Right: the same alarm explained in the game.

Why public understanding matters

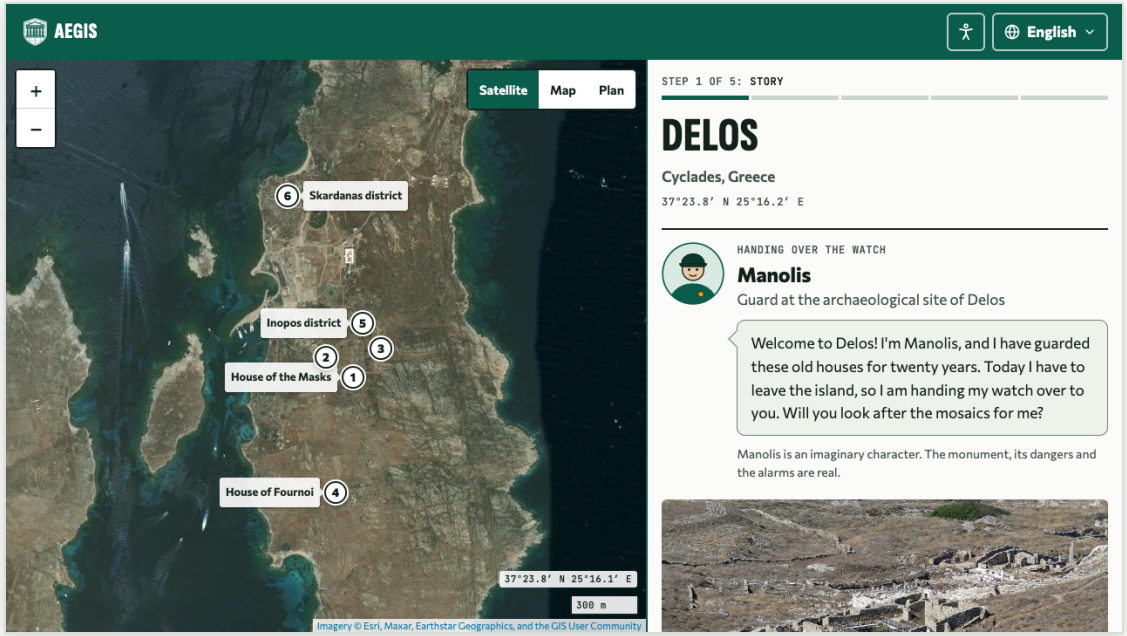
- Damage to a monument is slow and often unnoticed. Moisture weakens walls, sea salt erodes stone, and mosaics gradually lose their tesserae.
- By the time damage is visible, harm has already occurred.
- The knowledge behind prevention stays with specialists.
- Visitors see the outcome of preservation, but little of the work behind it. They may struggle to see how their own actions can help.



The game compares “Today” with “Years later, if nobody notices”. Drawings illustrate deterioration and its possible consequences. The game labels these images as illustrations of heritage risks, rather than photographs of actual damage.

Our idea: guardian for a day

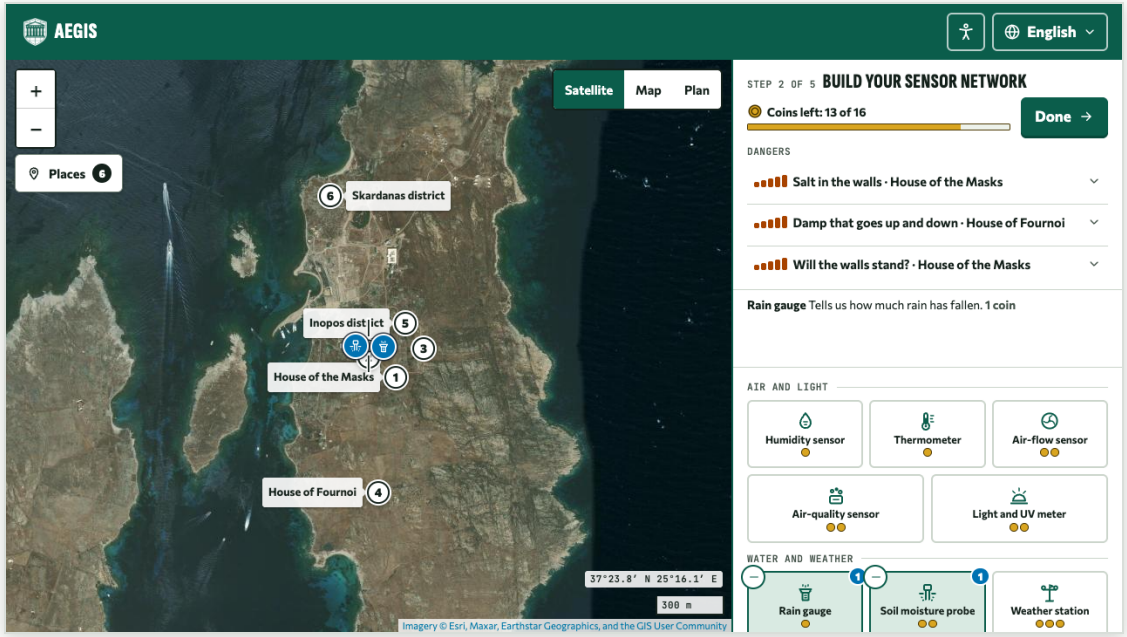
- An interactive experience makes the work of ARGUS easier to relate to than technical descriptions alone.
- In AEGIS, anyone can become a guardian at one of the five ARGUS sites, for up to seven days.
- A fictional guardian, such as Manolis on Delos, takes a day off and hands the watch to the player.
- The game makes the distinction explicit: the guardian is fictional, while the risks and recorded alarms come from real ARGUS monitoring data.



Gameplay: a sensor network and a real alarm

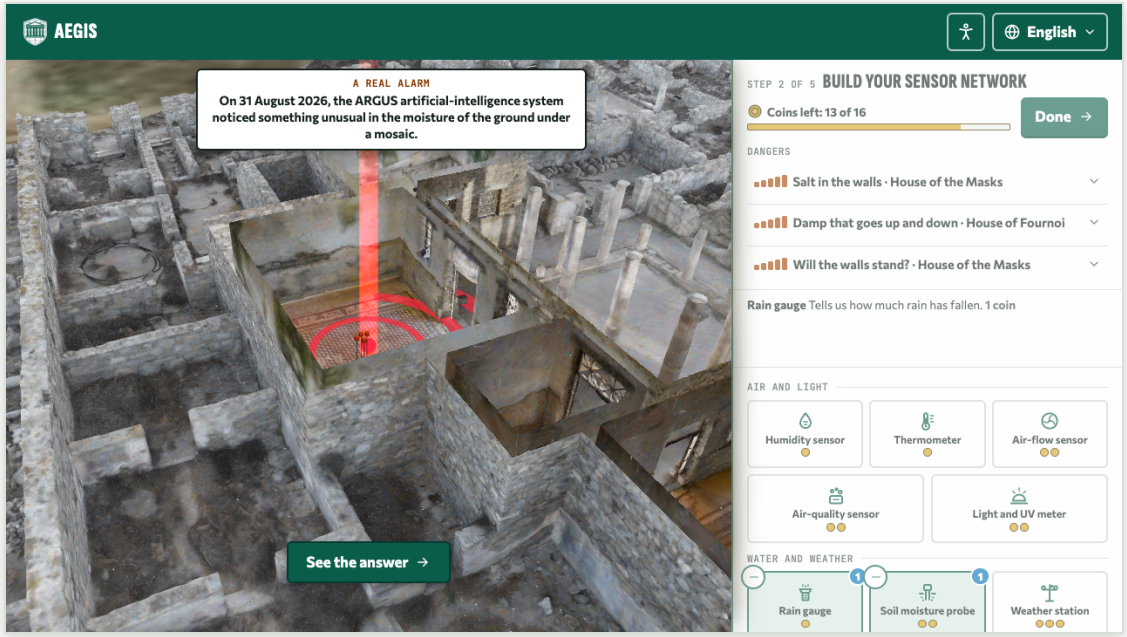
1. Building a sensor network

Players have 12–16 coins, depending on the risk level. They choose suitable sensors and position them on the monument to address three real threats.



2. Replaying a real alarm

Players encounter a recorded ARGUS alarm, such as the one raised on 31 August 2026. It appears at the location of the sensor that triggered it, on an ARGUS 3D model where one is available.



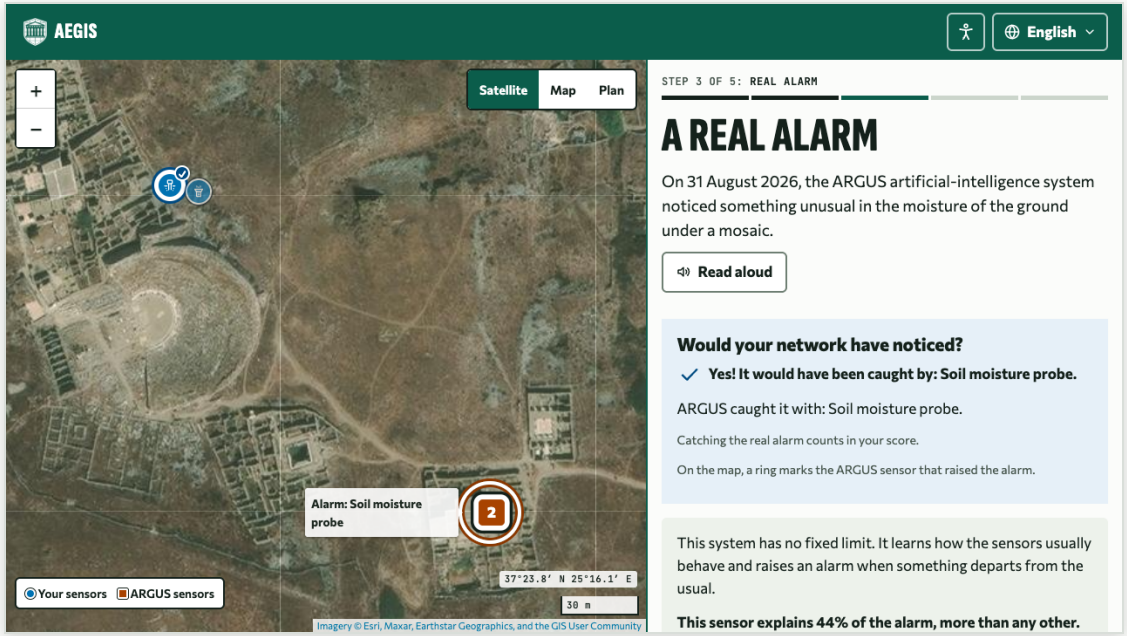
Gameplay: detection and comparison with ARGUS

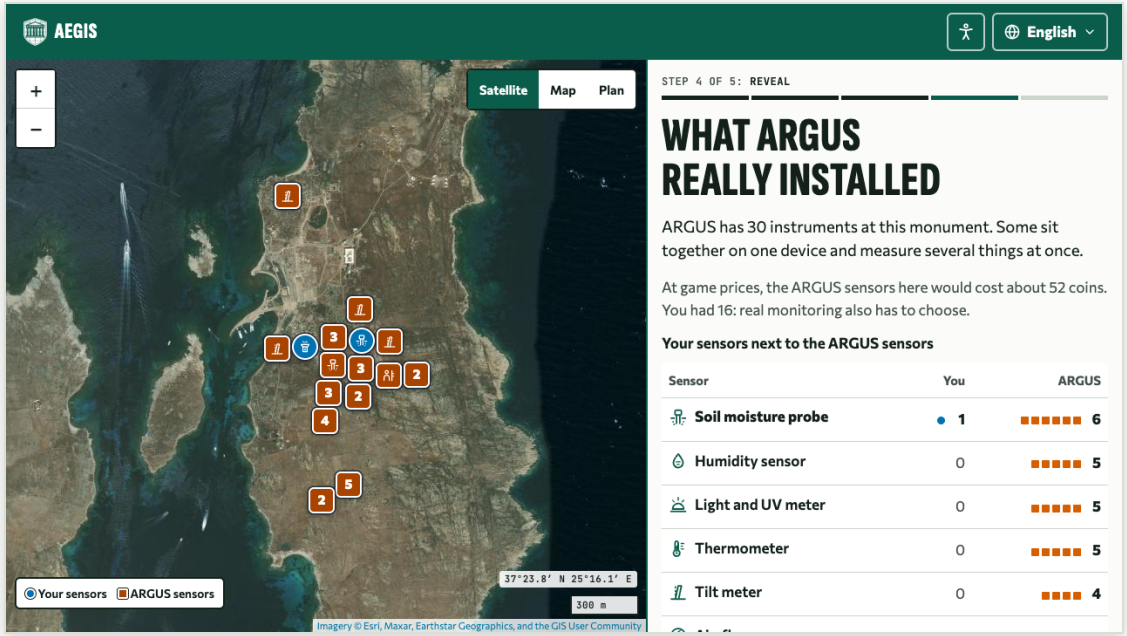
3. Would your network have detected it?

The game identifies the sensor that raised the alarm and shows how often it triggered. Players see whether their own network would have detected the event.

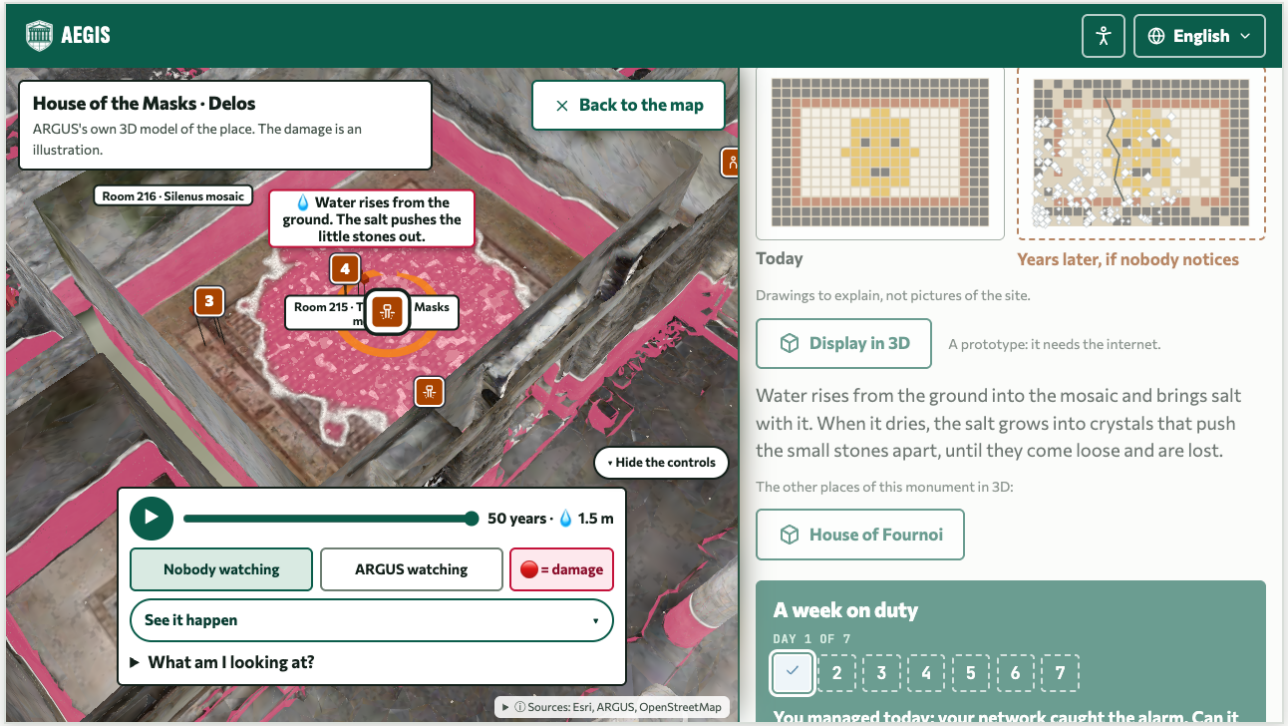
4. What ARGUS installed, and where

Players compare their network with the 30 instruments at Delos, shown in their actual positions. In game terms, the ARGUS network would cost 52 coins, prompting discussion of what to monitor and where.



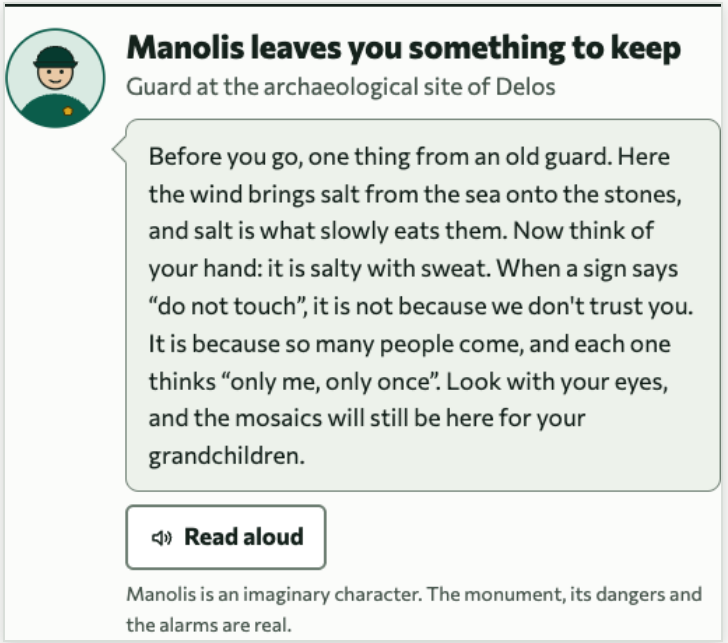


If nobody noticed



Up to 50 illustrative years without care, on the ARGUS 3D model of the House of the Masks. Water rises from the ground, salts crystallise, and mosaic tesserae loosen and disappear. The interface clearly labels these effects as educational illustrations rather than predictions.

The guardian returns and explains why visitors should follow the rules at a heritage site, even when damage is not yet visible.



How AEGIS uses ARGUS resources

The game draws its risks, installed sensor positions, and recorded alarms from ARGUS data across all five heritage sites.
The table connects those resources to what the player experiences.

ARGUS resource	Use in AEGIS
Coordinates of installed instruments	ARGUS's real network revealed on the map
Risk scenarios and daily risk assessments	The three threats in each mission
Alarms from August and September 2026, based on expert thresholds and AI models	Recorded alarms replayed for the player
Photogrammetric 3D models and terrain	Alarm locations and illustrated deterioration in 3D
Read-only API access, every two hours	"Live from ARGUS" on the opening screen
Heritage documentation structure (CIDOC CRM E14)	The visitor's structured observation report

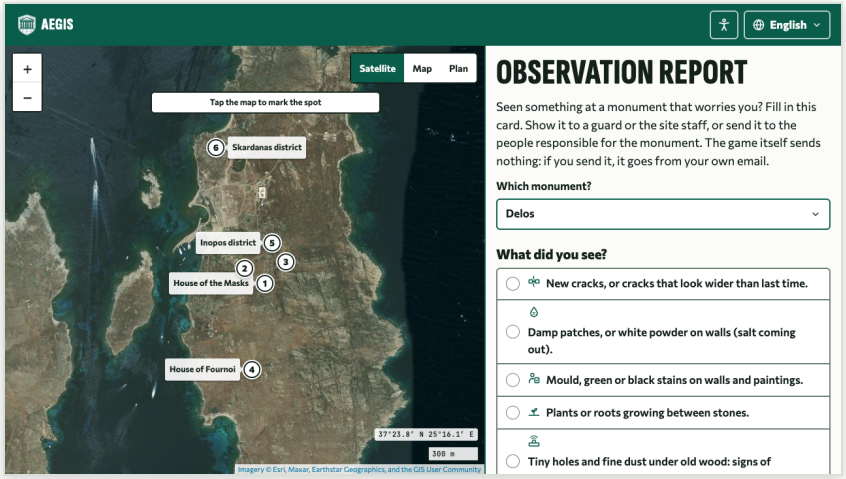
Benefits for visitors and ARGUS

For the visitor

- An explanation of each site's risks and preservation needs.
- A visitor code and a checklist of things to watch for on the next visit.
- A way to record an observation and, if desired, share it with the people responsible for the site.
- Teachers can compare class choices with the actual ARGUS network.

For ARGUS

- A public interface for its monitoring data.
- A learning resource built around those data.
- Citizen observations in its documentation structure, ready for possible future acceptance by ARGUS.



A report records what the visitor saw, where, and when.
The user decides whether to share it with a custodian.
The game sends nothing automatically.

Access and availability

- Five languages: Greek, English, Italian, Spanish, and German.
- Audio narration of on-screen text.
- Desktop and mobile browsers, with support for classroom use.
- A guided introduction on the first visit.
- Designed with WCAG 2.2 Level AA accessibility as a target.

Play the game:
`https://aloualou56.github.io/argus_trackA/`

